Integrating
Open OnDemand & Open XDMoD

Tutorial presented at
Global OnDemand Conference (GOOD) 2025

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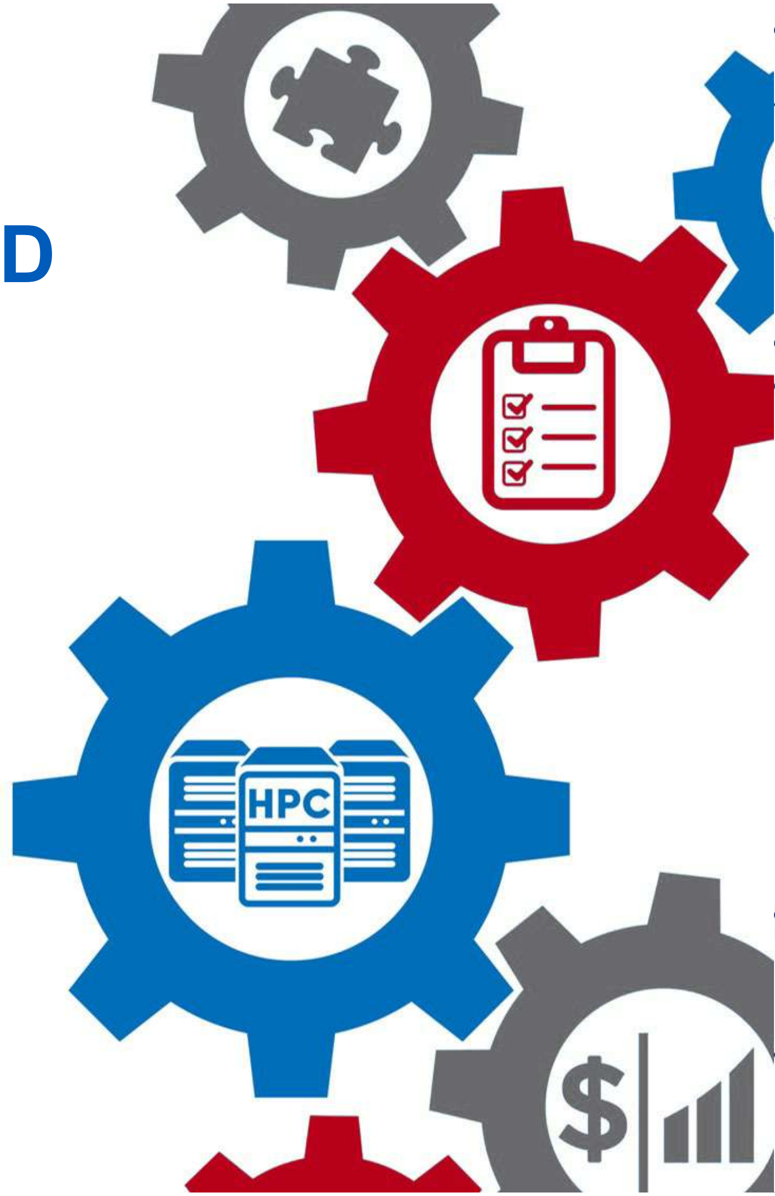
University at Buffalo Center for Computational Research



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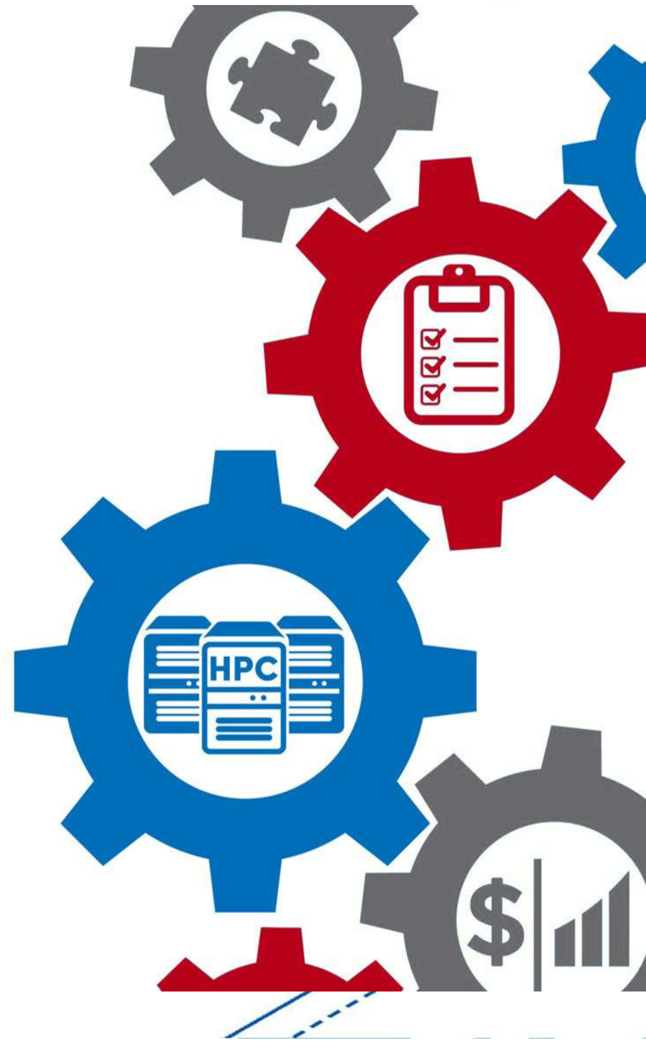
Agenda

- Tutorial technology & walk through of the repo
- A bit about XDMoD
- XDMoD integration with OnDemand
- OnDemand integration with XDMoD
- Demo/tutorial
- Where to find us/get help/stay connected
- Q&A



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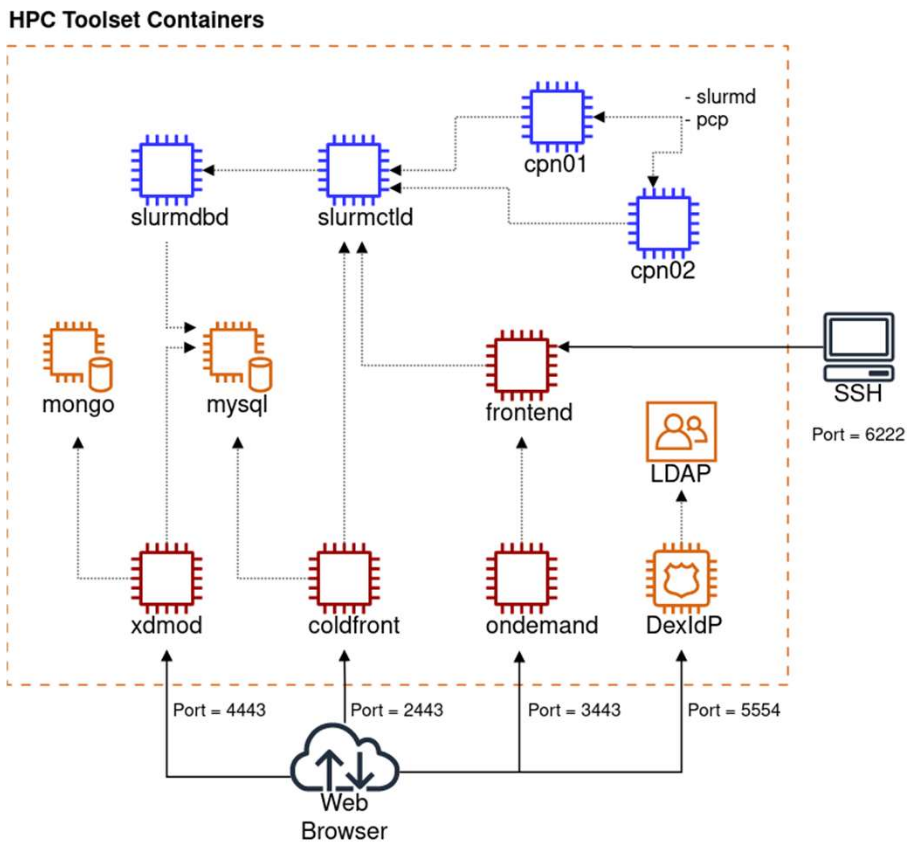
Tutorial Container Architecture



Requirements: https://github.com/ubccr/hpc-toolset-tutorial/edit/master/docs/requirements.md

Clone the Github Repo:

```
git clone https://github.com/ubccr/hpc-toolset-tutorial
cd hpc-toolset-tutorial
./hpcts start
```

* The first time you do this, you'll be download ~25GB worth of containers from Docker Hub. This can take a long time depending on your network speeds. After downloaded, the containers are started, and services launched.

WARNING!!! DO NOT run these containers on production systems. This project is for educational purposes only. The container images we publish for the tutorial are configured with hard coded, insecure passwords and should be run locally in development for testing and learning only.

HPC Toolset Tutorial Walk Through

https://github.com/ubccr/hpc-toolset-tutorial

Keep the applications page open for easy access to account usernames/passwords and portal URLs:

https://github.com/ubccr/hpc-toolset-tutorial/blob/master/docs/applications.md

We'll be following the XDMoD-OnDemand Integration Tutorial :

https://github.com/ubccr/hpc-toolset-tutorial/blob/master/xdmod/README.md

If you're following along, please do the first step of the tutorial now:

[hpcadmin@frontend ~]$ submit_jobs.sh

XDMoD: Metrics on Demand

- **Comprehensive framework for CI system management**
  - HPC Jobs, Storage, Cloud, Networking, OnDemand, & other Gateway data
- **Understand and optimize resource utilization and performance**
  - Provide instantaneous and historical information on utilization
  - Measure Quality of Service of CI systems and applications
  - Measure and improve job and system level performance
  - Inform computing system upgrades and procurements
- **ACCESS XDMoD Tool**
  - Analytics Framework for ACCESS and NAIRR
- **Open XDMoD*: Open Source version for CI centers**
  - Used to measure and optimize performance of CI centers
  - 500+ academic, governmental, & commercial installations worldwide

https://open.xdmod.org/

XDMoD
METRICS ON DEMAND

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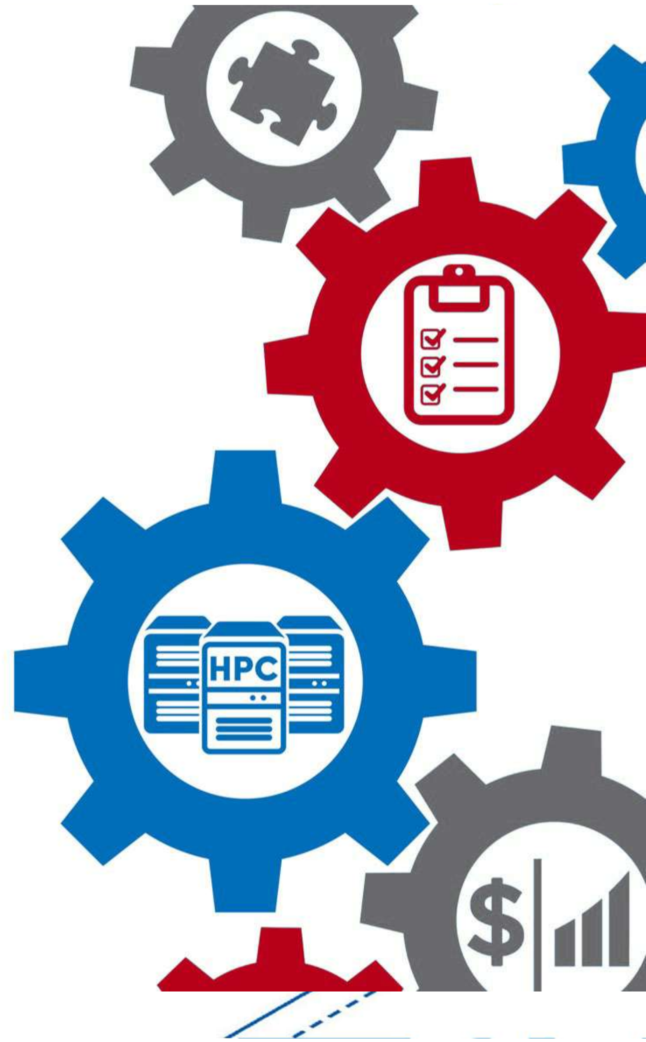
Open XDMoD – Usage & Performance Metrics

- Tool that aggregates job data & system performance metrics to inform system users, system staff & center decision makers
- Web portal providing job & system metrics, including utilization, quality of service metrics designed to proactively identify underperforming system hardware and software, and job level performance data for every job
- Role-based access to data with different levels of granularity, including job, user, or on a system-wide basis
- Ingest OnDemand logs into new OnDemand realm in XDMoD



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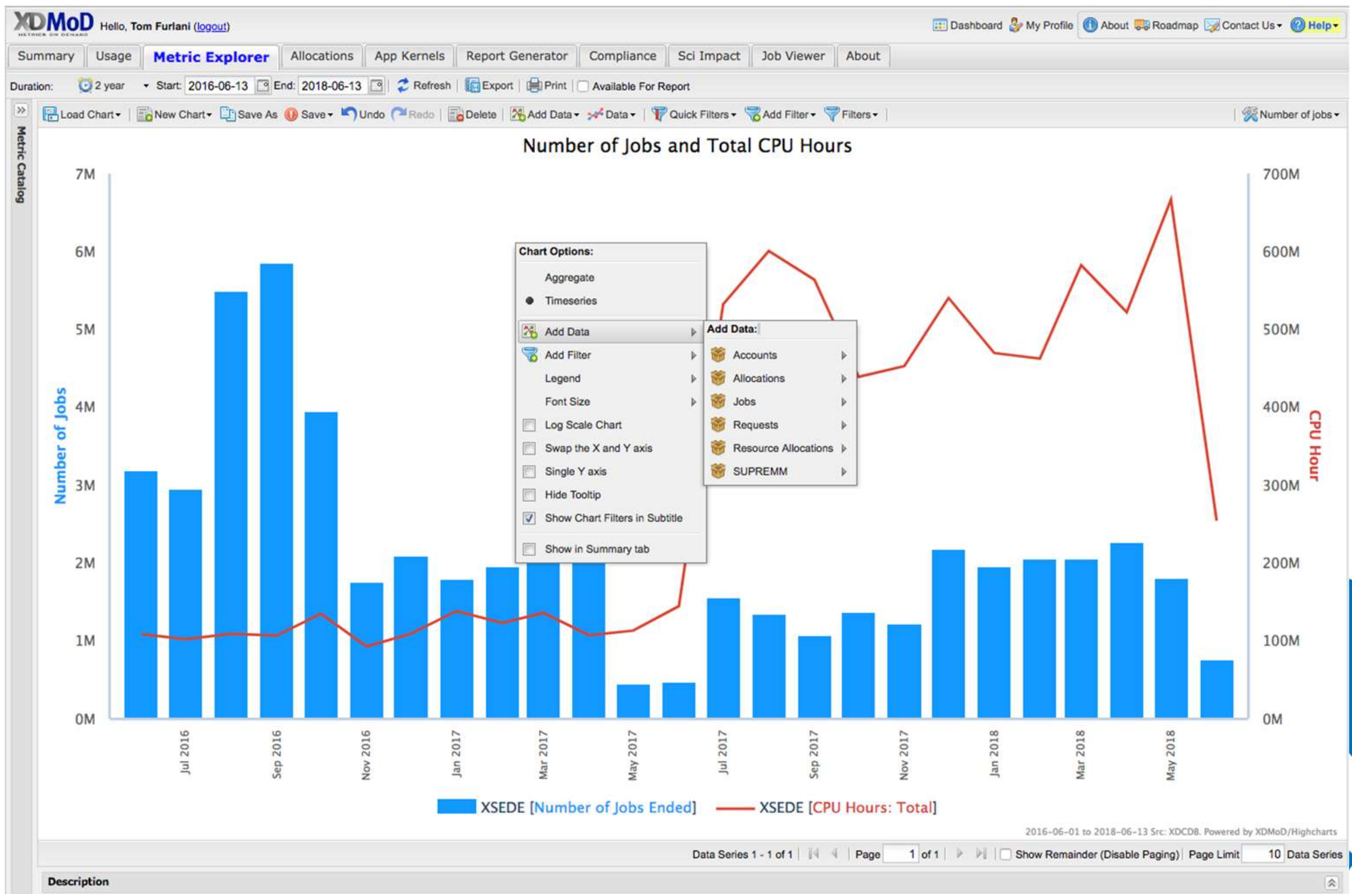
ACCESS Metrics

XDMoD Portal

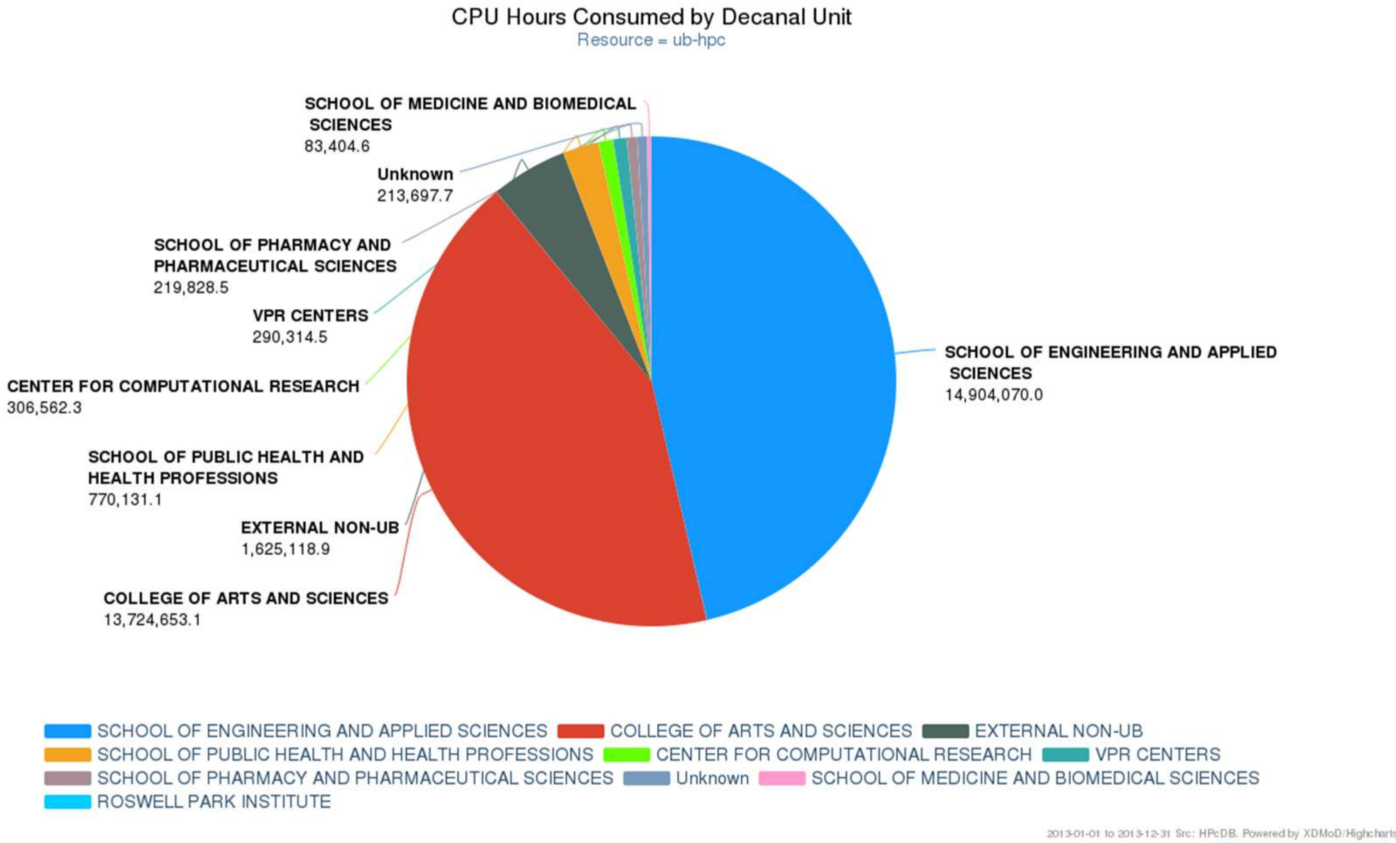
- **Web-based**
  - Point and click drill down capability
- **Display metrics**
  - Utilization, performance, scientific impact
- **Role based access**
  - User
  - Principal Investigator
  - Center Staff
  - Center Director
- **Custom Report Builder**



XDMoD
METRICS ON DEMAND

Easily Obtain Utilization Metrics

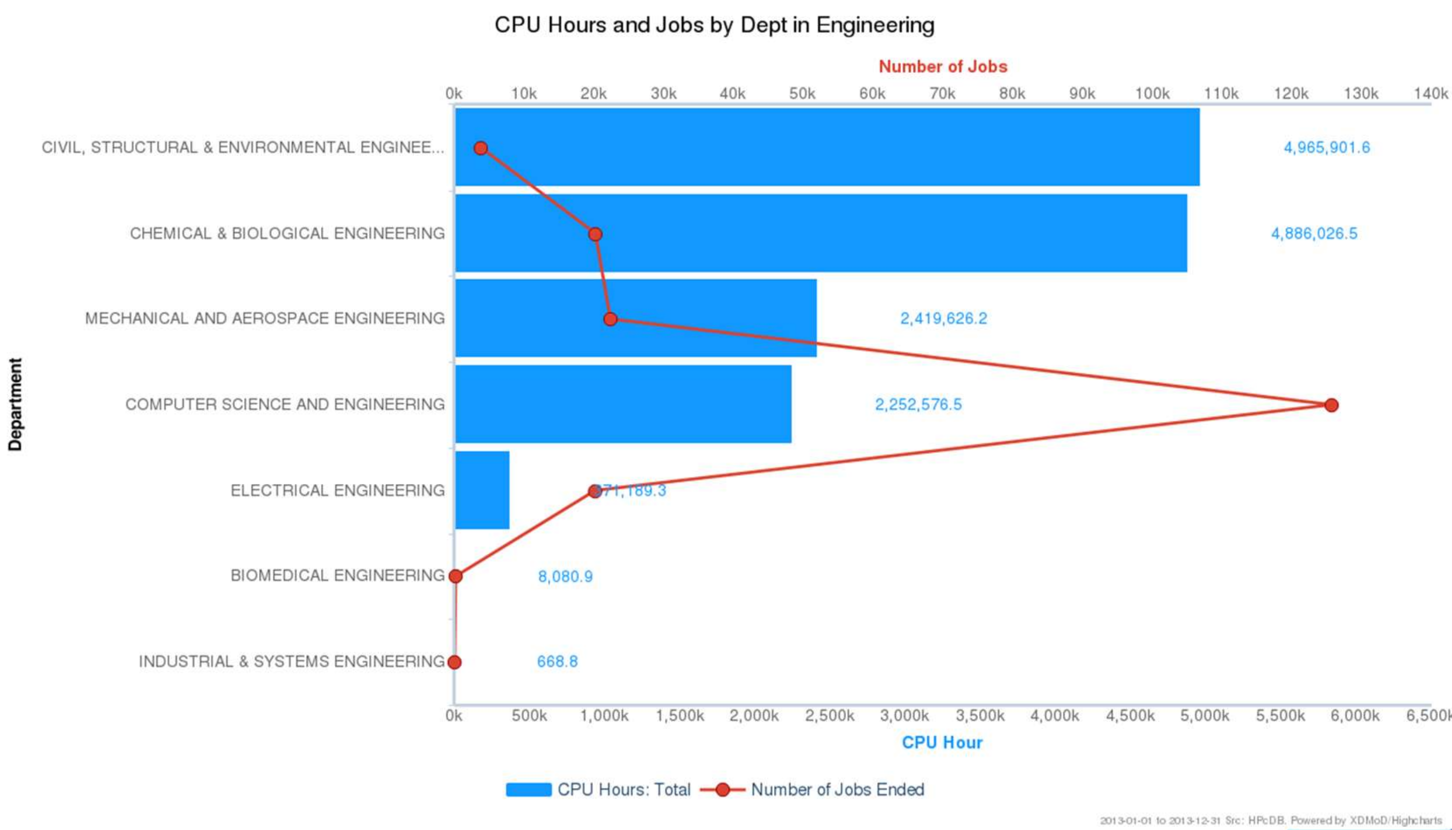
CPU hours consumed by campus units



XDMoD
METRICS ON DEMAND

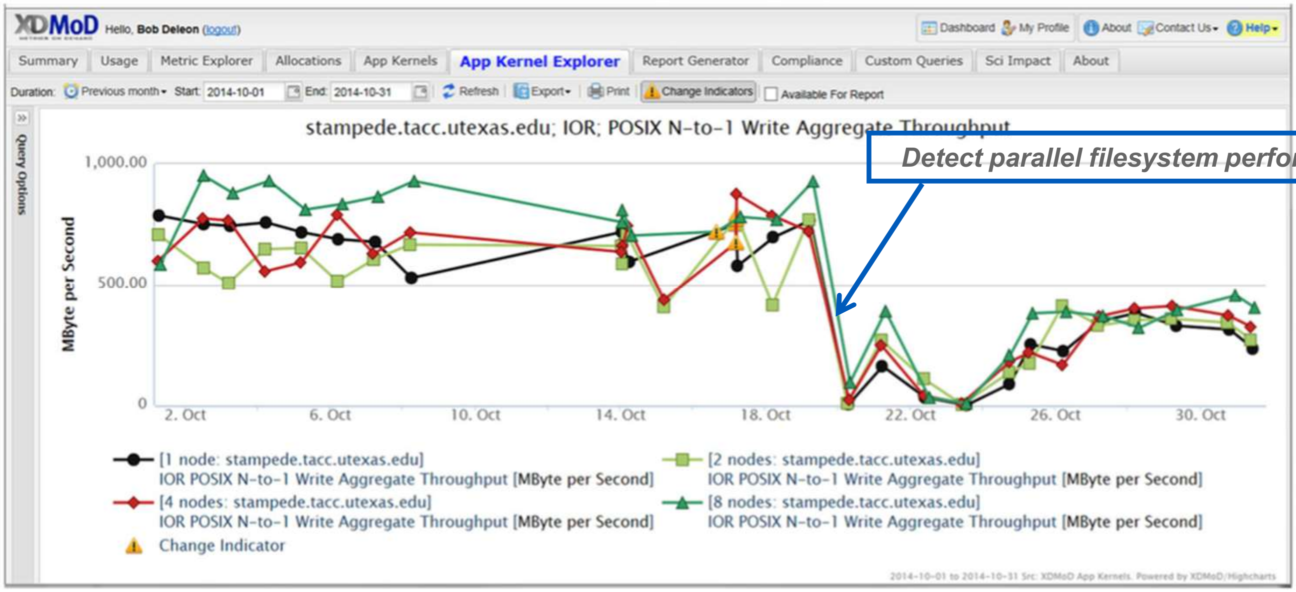
Drill Down for Greater Detail

CPU hours and jobs by Engineering Department



QoS: Application Kernels

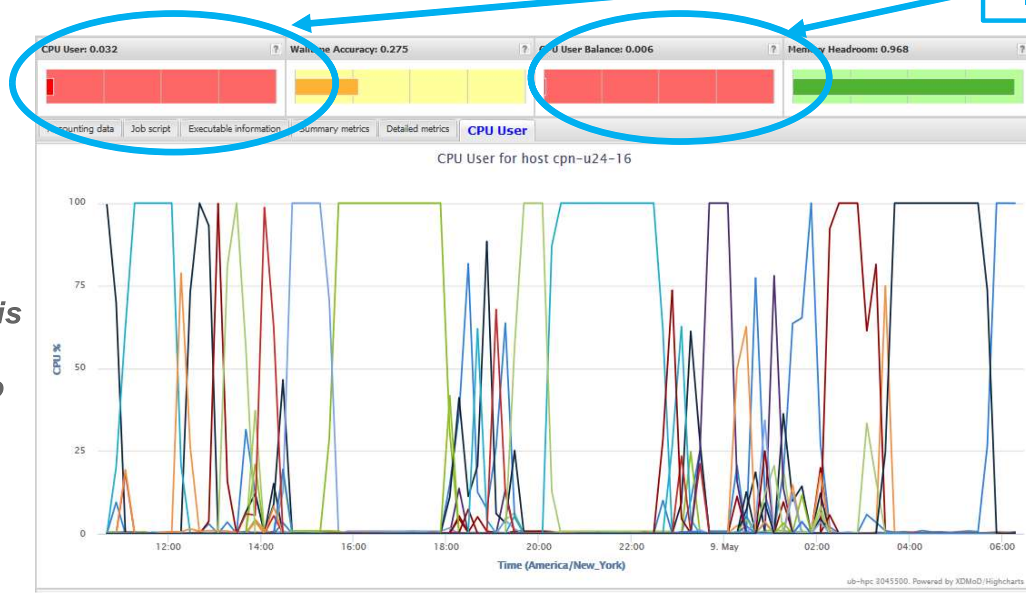
- **Computationally lightweight**
  - Run continuously and on demand to actively measure performance
- **Measure system performance from User's perspective**
  - Local scratch, global filesystem performance, local processor-memory bandwidth, allocatable shared memory, processing speed, network latency and bandwidth
- **Proactively identify underperforming hardware and software**



Measuring Job Level Performance

- **Detailed job-level performance data**
- **Run on every cluster node and collects hardware counter data**
  - CPU usage, Memory usage, I/O usage
- **Identify poorly performing jobs (users) and applications**
  - Support personnel/Users can use tools to identify/diagnose problems

"Traffic Lights"
Indicate poor core utilization

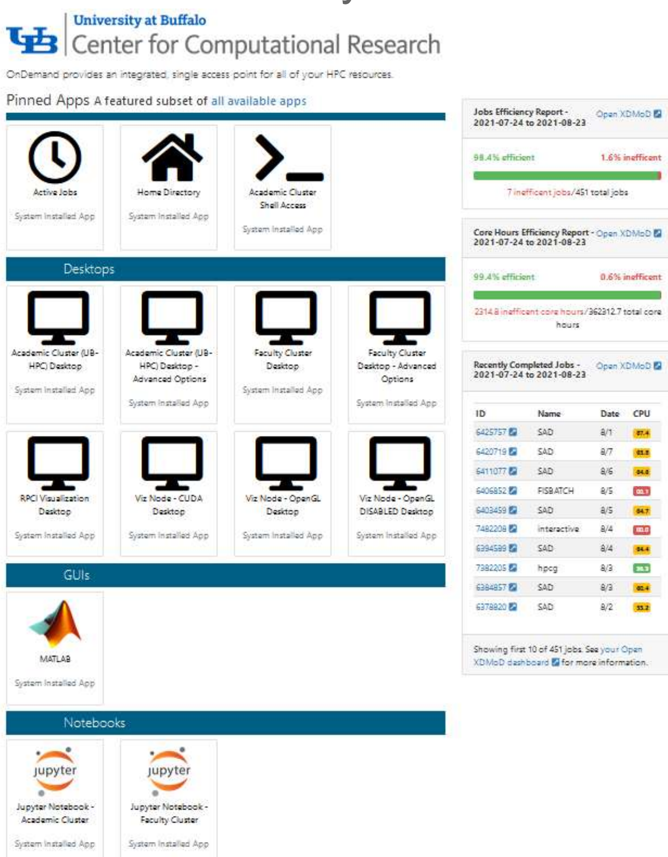


Job employs 32 cores but only 1 core is active at a time with control swapping between the 32 cores Minor change to the submission script resulted in an improvement of a factor of 30

XDMoD
METRICS ON DEMAND

Open XDMoD - OnDemand Integration

Provide XDMoD information directly to OnDemand users



Can access job viewer in a single click

Overall job efficiency summary

Core hours efficiency summary

Individual job level efficiency

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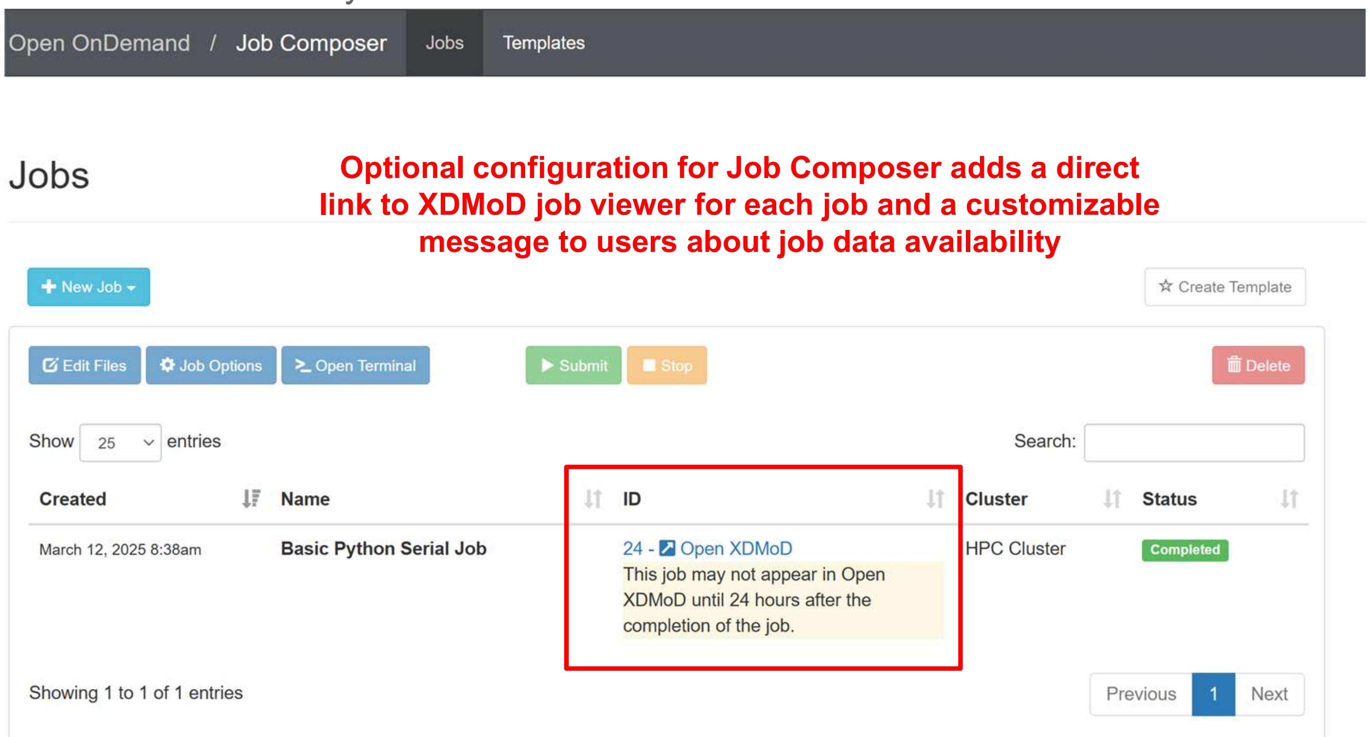
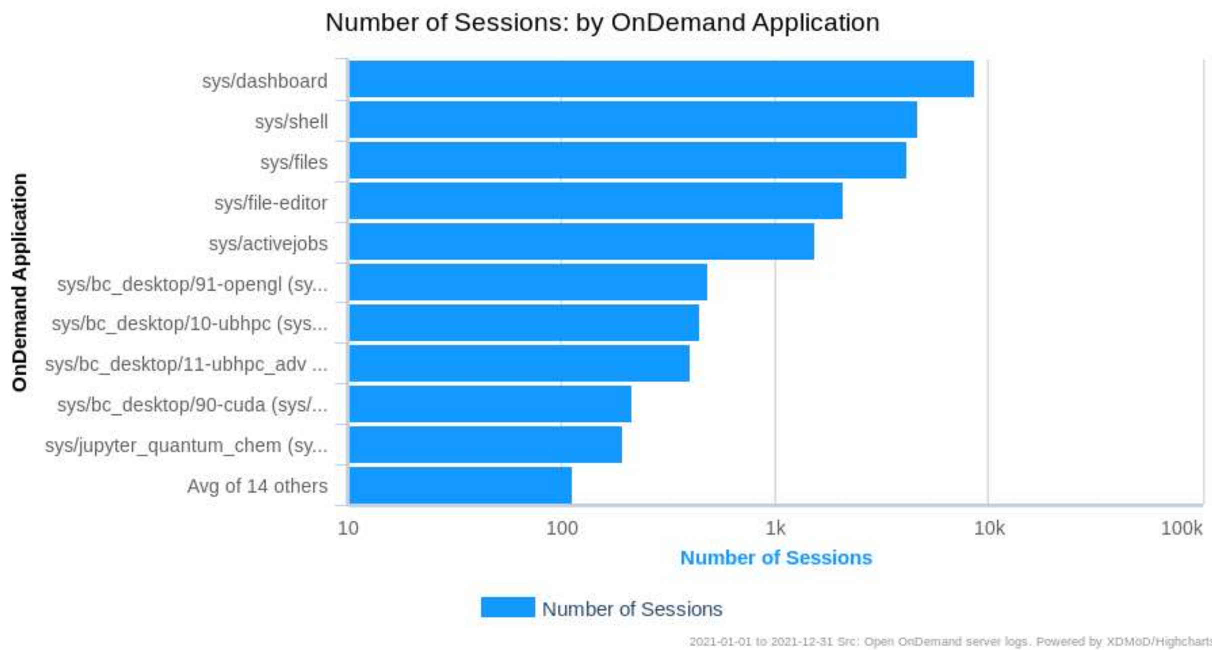


XDMoD
METRICS ON DEMAND

OnDemand – Open XDMoD Integration

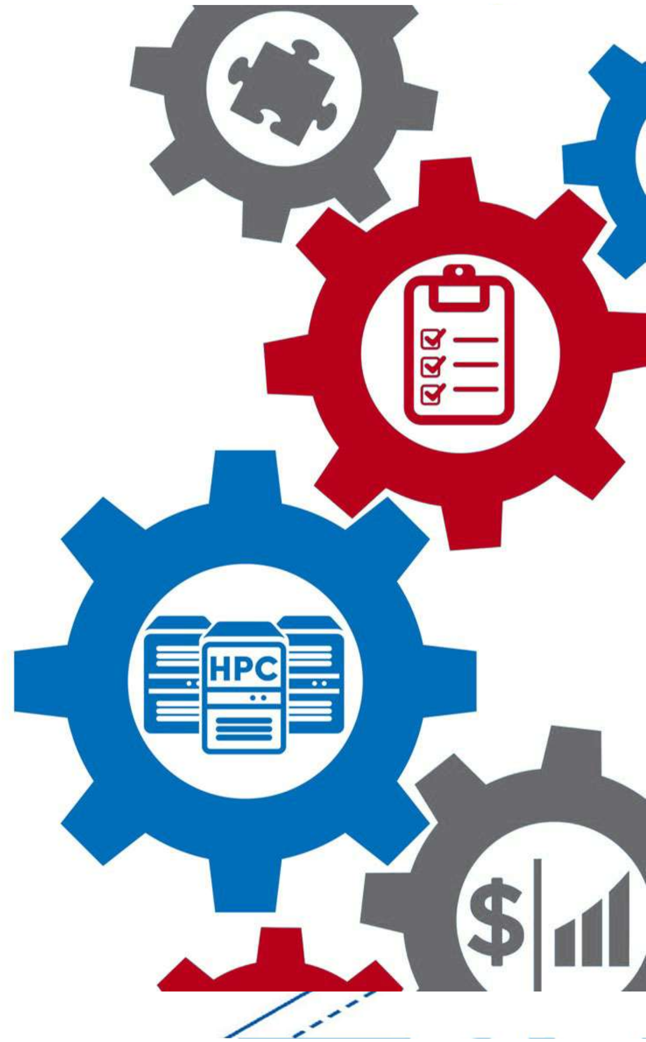
View OnDemand usage in XDMoD

- Who is using OnDemand?
- What are they running?
- Where are they running from?



ColdFront – Managing Access

- Used as the source of record in an HPC center to ensure security & continuity of the systems
- Provides center staff ability to manage center resources & who has access to them
- Portal for users to manage their access to center resources & report on their research
- Plug-ins for job scheduler (Slurm), central authentication, job statistics (XDMoD), OnDemand, that enable automation of access to or removal from resources
- Reports for center management to demonstrate the center’s impact (publications, grants, research output)



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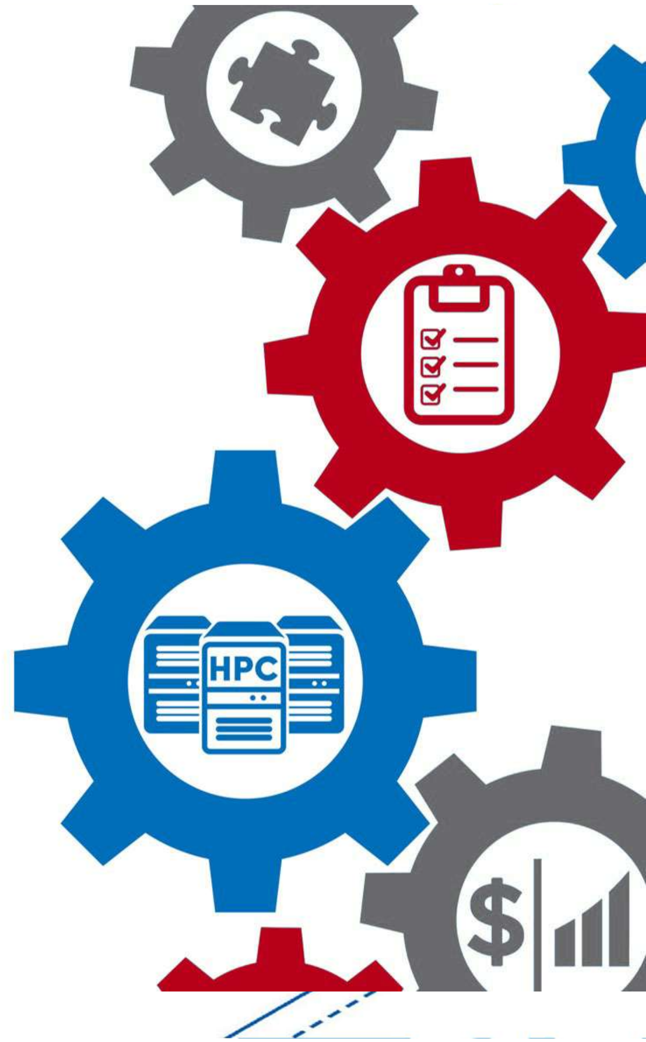

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- We gratefully acknowledge the partnership with Virginia Tech on the previous OnDemand2.0 NSF project

How to reach us:

Center for Computational Research – https://buffalo.edu/ccr

Open XDMoD - https://open.xdmod.org/

ColdFront – https://coldfront.io

Connect with us!

Join the Slack organization for the tutorial

https://tinyurl.com/hpctoolset